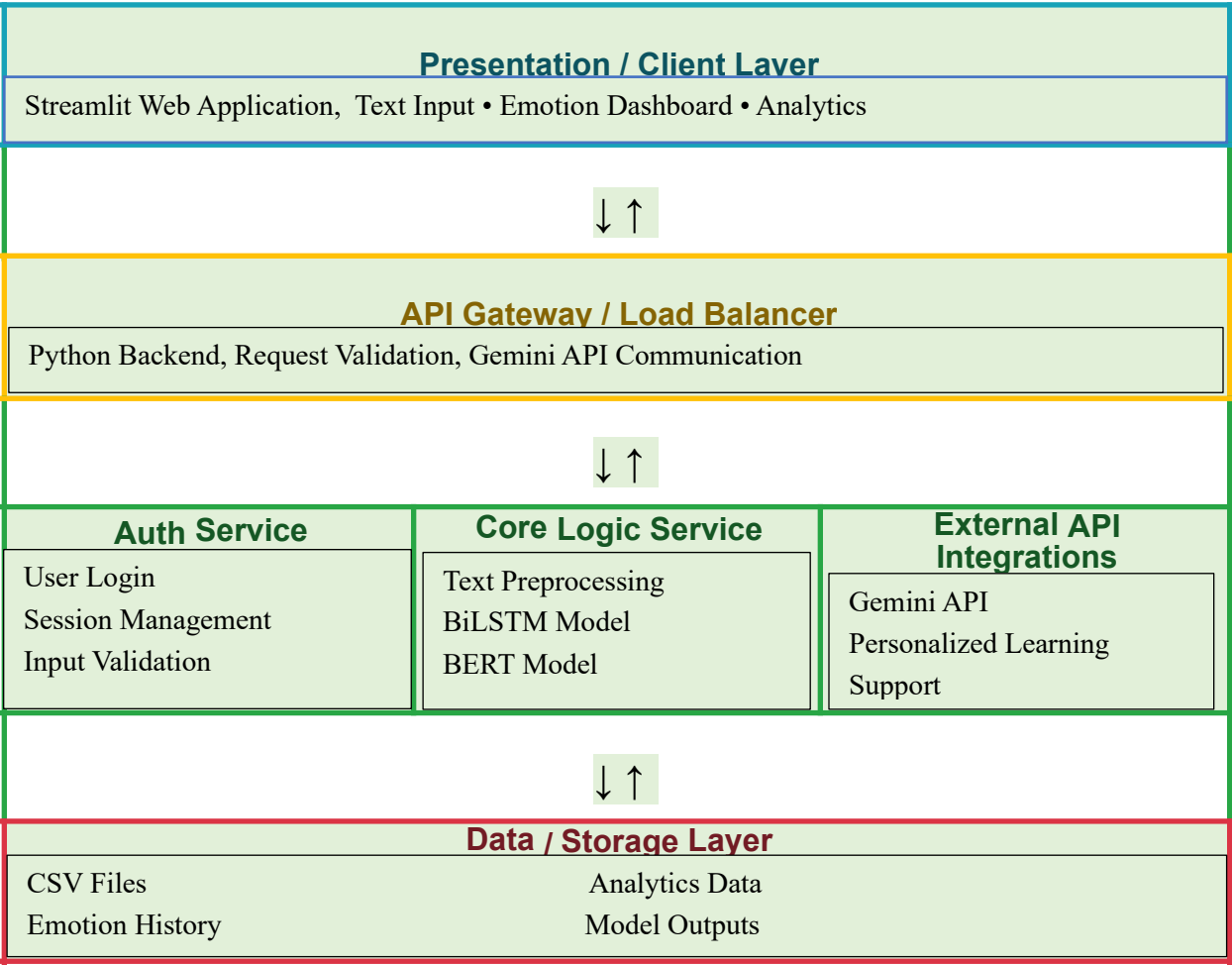


Solution Architecture

Date	15 March 2026
Team ID	xxxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

The **Emotion Detection & Learning Support Engine** follows a three-layer architecture consisting of the Presentation Layer, Application Layer, and Data Layer. Users interact with the Streamlit web interface to submit learning-related text. The application processes the input using NLP techniques, predicts emotions through BiLSTM and BERT models, generates personalized learning guidance using the Gemini API, and stores the results for visualization and analysis.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for text input, emotion visualization, recommendations, and analytics.	Streamlit
API Gateway	Validates requests and connects the frontend with AI models and external APIs.	Python, Streamlit
Microservices	Performs preprocessing, emotion detection, confidence scoring, and AI recommendation generation.	BiLSTM, BERT, TensorFlow, Transformers, NLTK, Gemini API
Database	Stores user inputs, emotion history, prediction results, and analytics.	CSV, Pandas